



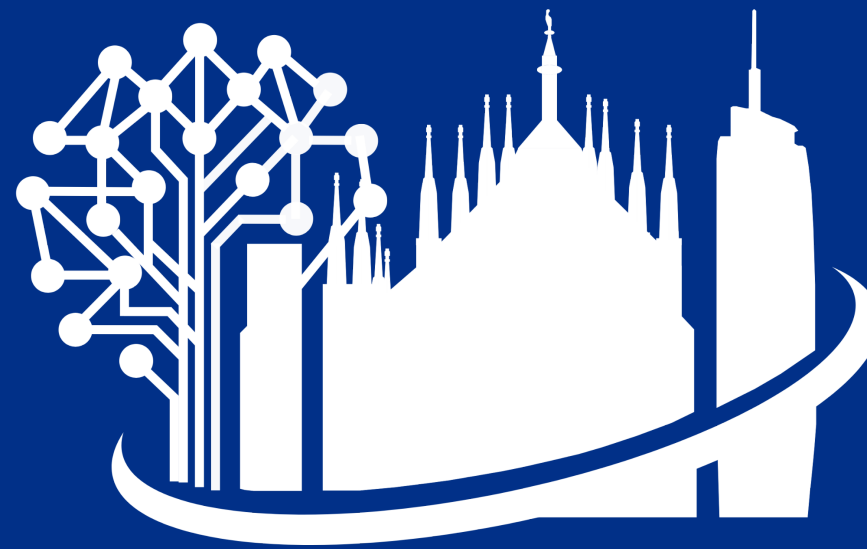
Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

A portfolio allocator trained by reinforcement learning (RL) to maximise return has no incentive to limit large losses. The usual safeguard caps Conditional Value-at-Risk (CVaR), the mean loss in the worst weeks, through a Lagrangian penalty. We identify two ways in which this safeguard fails without visible symptoms. When the penalty and the reward are on different scales, the multiplier stays near zero and the budget is never enforced, while tail metrics still improve through other parts of the pipeline. The evaluation protocol can also decide the ranking. On a 31-asset universe, a single stress window placed the learned allocator above every classical optimiser, and a rolling walk-forward reversed the order. We add a decision-time filter that forecasts the CVaR of each proposed portfolio from trailing data and shrinks it toward minimum variance when the forecast exceeds the budget. At matched settings the filter lowered the out-of-sample breach rate from **75.1% to 7.4%**. Over a 40-fold walk-forward on two ETF universes, the constraint reduced CVaR₉₉ in **26 of 27** untied folds (Wilcoxon $p = 8.8 \times 10^{-6}$), and the result held after Benjamini-Hochberg correction and calendar clustering.

1 Introduction

In the week of 16 March 2020 the S&P 500 fell about 15% over five sessions and the VIX closed at 82.69. An allocator trained without a loss limit continued to hold equities. An otherwise identical allocator with a CVaR budget had moved into Treasuries and cash before the trough, and its drawdown was 7.7% against 16.3% (Fig. 1). Over 2020 to 2024 the maximum drawdowns were 11.8% and 20.6%.

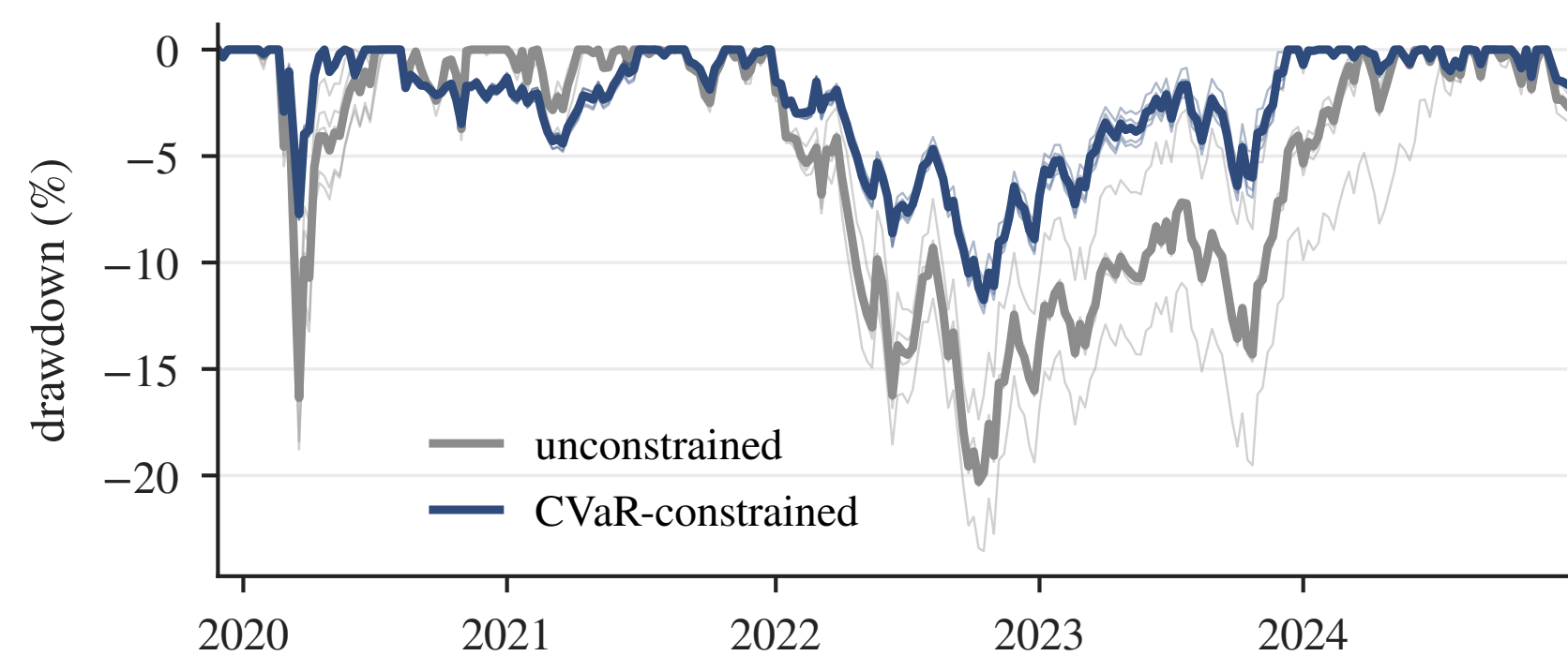


Figure 1. Drawdown on the 2020 to 2024 stress window. Thick lines follow the five-seed mean return path and thin lines the individual seeds.

The standard way to impose such a limit in RL is a Lagrangian penalty [2] whose multiplier λ rises while the budget is exceeded. Safe-RL work has reported that λ responds slowly and depends on its step size [3], and OmniSafe standardises the cost signal by default [4]. Portfolio RL studies usually report a single test window, although optimised portfolios that look strong in one sample often fail out of sample [6, 7]. The poster makes three contributions.

1. Two failure modes with diagnostics, a Lagrangian constraint-coupling failure and an evaluation-protocol reversal.
2. A differentiable CVaR-constrained allocator with a decision-time filter that reduced out-of-sample breach from 75.1% to 7.4% at matched dual settings.
3. A two-universe rolling walk-forward with Wilcoxon, Benjamini-Hochberg and Deflated Sharpe controls.

2 Problem formulation and data

Each week t the agent observes s_t (26-week momentum and volatility per asset, current weights, drawdown and a rolling 95% CVaR estimate) and chooses post-trade weights w_t on the long-only simplex. The environment projects w_t onto the admissible set (maximum weight 0.40, turnover cap 0.50) and realises $r_t = w_t^\top \rho_t - c \|w_t - w_{t-1}\|_1$ with $c = 5$ bps. Features at t use only returns before t . The problem is a constrained MDP [2],

$$\max_{\pi} J(\pi) \text{ s.t. } \text{CVaR}_{\alpha}(L) \leq d, \quad \text{CVaR}_{\alpha}(L) = \min_{\zeta} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[(L - \zeta)^+] \right\} [1],$$

with $\alpha = 0.95$ and $d = 1.2\%$. A portfolio is over budget when its CVaR_{0.95} over the trailing 52 realised weeks exceeds d . The breach rate is the share of test weeks over budget, with 5% tolerated. Most results report CVaR_{0.99}, a deeper tail than the one constrained.

The primary universe has seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL) with 887 weekly returns from 2008 to 2024. The stress study trains on 2008 to 2018 and tests on the 267 weeks that contain the 2020 crash and the 2022 selloff. A ten-sector ETF universe and a 31-asset cross-asset universe use the same source, clock and costs. Seeds are 7, 13, 23, 42 and 2025.

3 Method

The actor is an MLP whose output tilts an adaptive inverse-volatility anchor a_t , so that $w_t = \text{softmax}(\log a_t + f_{\theta}(s_t))$ starts at the anchor. Over randomised training windows we back-propagate

$$\mathcal{L} = -\text{Sharpe} + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad \lambda \leftarrow \max\{0, \lambda + \eta_{\lambda}(\widehat{\text{CVaR}} - d)\},$$

where $\widehat{\text{CVaR}}$ is the mean loss over the worst $(1 - \alpha)$ fraction of window returns. Its $\text{top}k$ subgradient acts only on tail weeks. The selected checkpoint has the best validation objective among those within the validation CVaR limit.

Before each trade, the filter forecasts the proposal's CVaR from the trailing $L = 104$ weekly returns as $\widehat{\text{CVaR}}(w) = c_{\alpha} \sigma_p - \mu_p$, with $c_{\alpha} = \varphi(\Phi^{-1}(\alpha))/(1 - \alpha)$ and $c_{0.95} = 2.06$. If the forecast exceeds d it trades

$$w(s^*) = s^* w + (1 - s^*) w^{\text{MV}}, \quad s^* = \max\{s \in [0, 1] : \widehat{\text{CVaR}}(w(s)) \leq d\},$$

where w^{MV} is the trailing minimum-variance portfolio, following the safety-layer idea of [5]. The decision at t depends only on $\mathcal{F}_{t-1}$, and a shifted-returns test showed zero deviation.

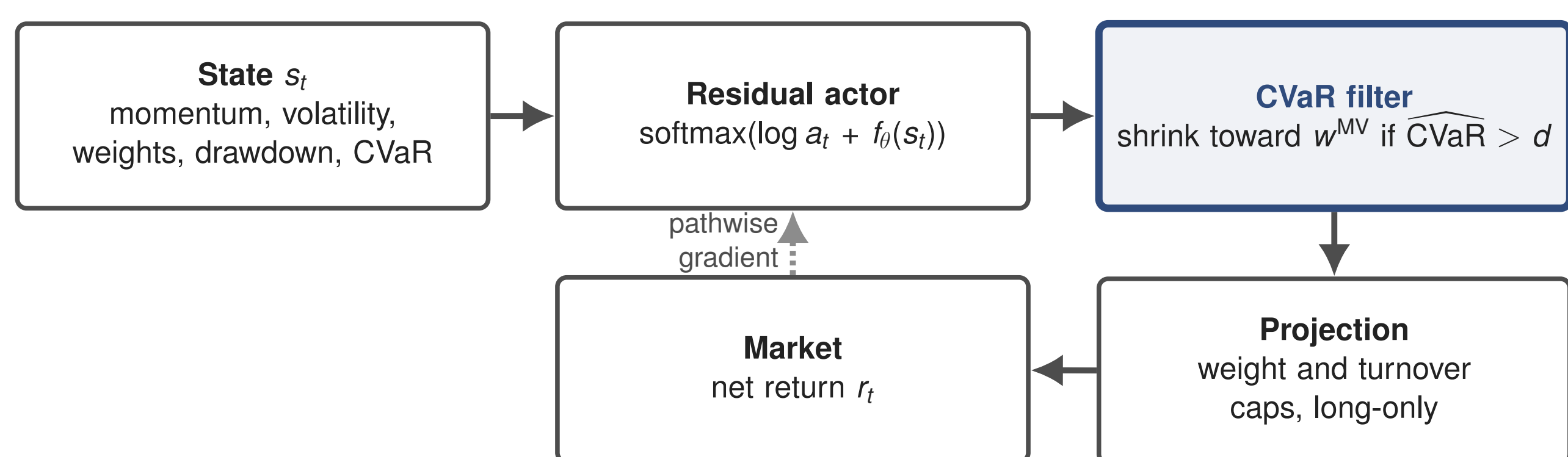


Figure 2. Allocation pipeline. The filter sits between the actor and the projection.

4 Evaluation-protocol reversal

On the 31-asset universe, a single stress split (training before 2018, testing through 2024) placed the constrained learner above every optimiser, with Sharpe 0.85 against 0.23 for minimum variance. Under a rolling walk-forward with 20 out-of-sample folds refit every 26 weeks, minimum variance led with 1.40 against 0.74 (Fig. 3). Covariance shrinkage did not account for the split result, since Ledoit-Wolf minimum variance [8] reached 0.25 and the min-CVaR LP 0.26 on the same split. The tail-risk reduction against unconstrained RL held under both protocols (fold-level Wilcoxon $p = 6 \times 10^{-4}$), whereas the Sharpe advantage over the optimiser appeared only in the single split.

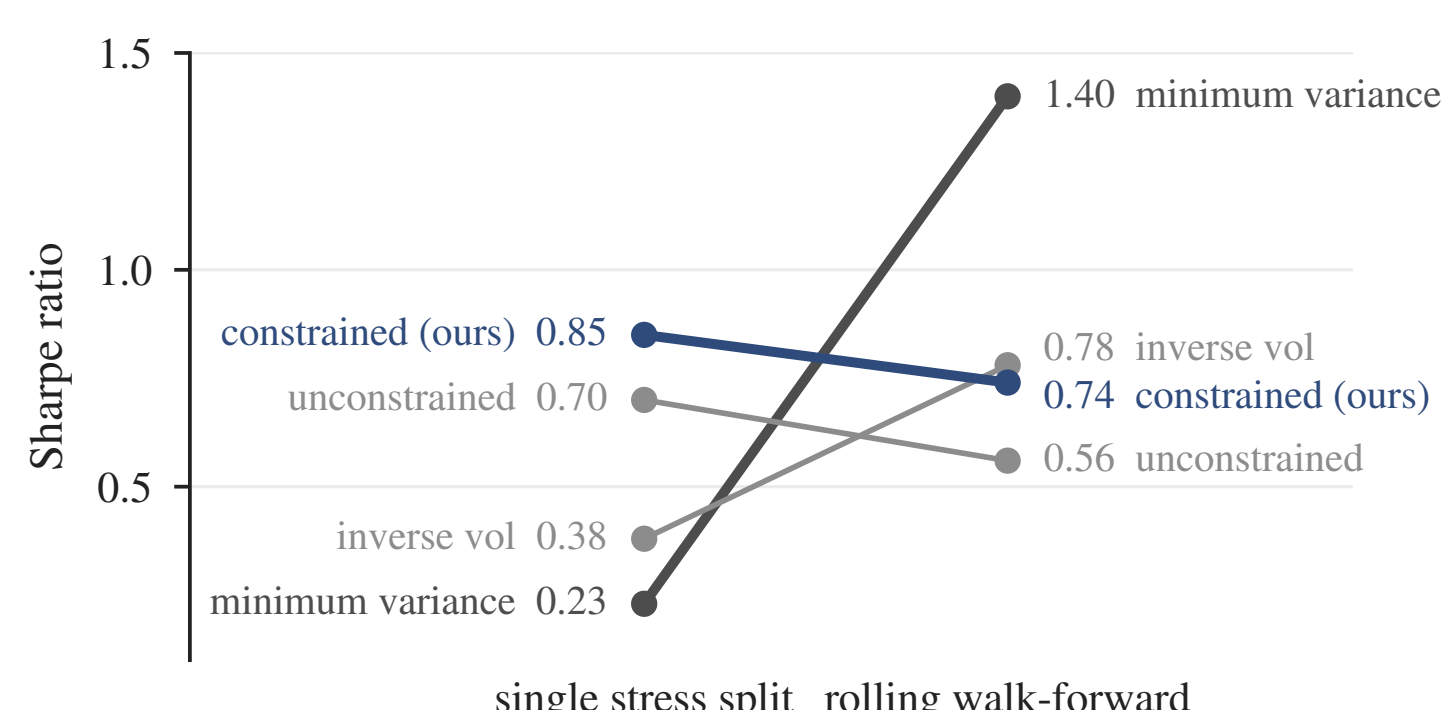


Figure 3. Sharpe ratio of four methods under two protocols, 31-asset universe.

Comparisons between learned allocators and optimisers should use rolling walk-forward folds with Deflated Sharpe [7]. Single stress-window comparisons are unreliable for this purpose.

5 Constraint-coupling failure

Four arms of the differentiable allocator share the actor, projection and budget (5 seeds, 1500 updates). They are A1 unconstrained, A2 mis-scaled dual ($\eta_{\lambda} = 0.001$), A3 scaled dual ($\eta_{\lambda} = 5.0$) and A4 scaled dual with a loose budget ($d = 1.92\%$). With the mis-scaled dual, λ stayed near zero for the whole run, with a mean final value of 0.022 against 0.101 for the scaled dual (Fig. 4). The CVaR excess is of order 10^{-2} in weekly-return units, so a step of 10^{-3} moves λ by about 10^{-5} per update. A model-free actor-critic failed in the same way. Its reward advantage was standardised while its cost advantage stayed near 10^{-3} , and the constrained and unconstrained agents trained identically until the cost advantage was also standardised.

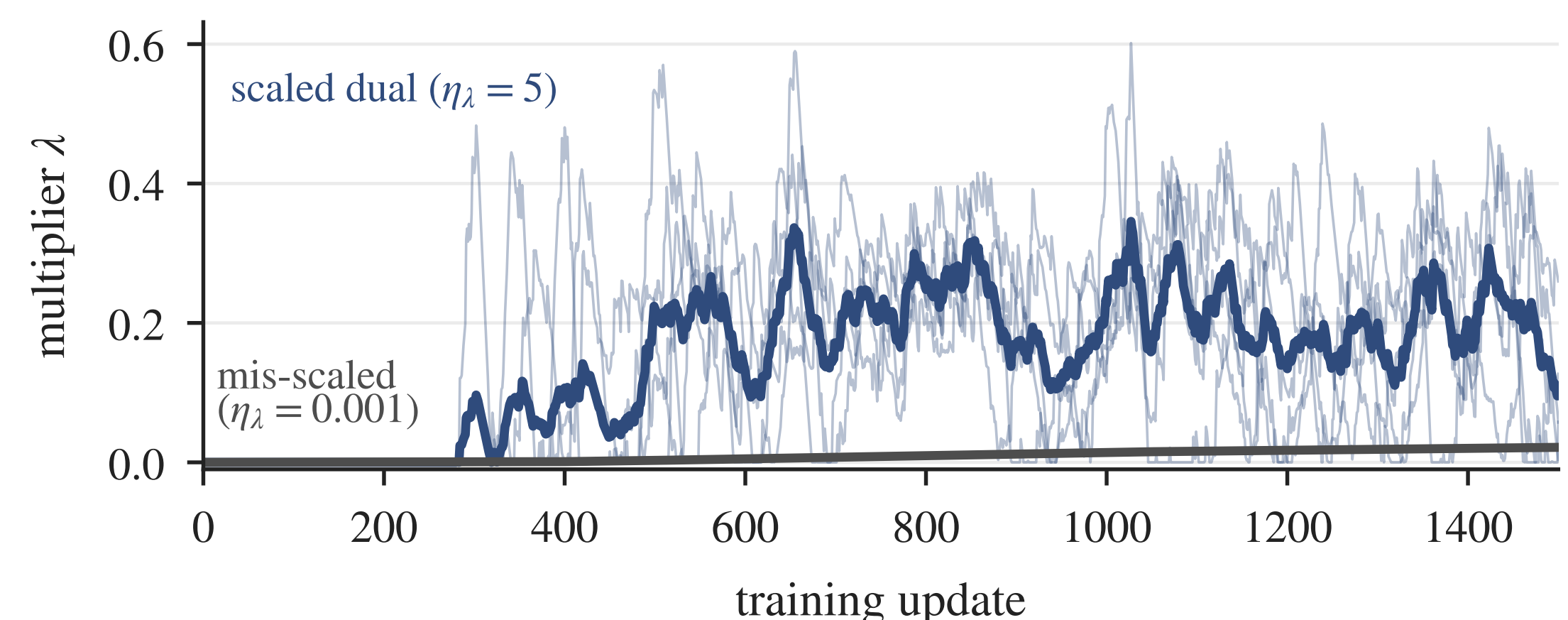


Figure 4. The Lagrange multiplier over training. Thick lines are five-seed means.

Tail metrics do not reveal the failure. Even with λ near zero, the constrained pipeline lowered CVaR₉₉ by 61% relative to A1 (0.025 against 0.065) and raised Sharpe from 0.626 to 0.987 (permutation test, one-sided $p = 0.0044$, $n = 50,000$). The reduction came from checkpoint selection under the CVaR limit together with the shared anchor and projection. Engaging the dual changed CVaR₉₉ little (0.028 against 0.025).

A multiplier that remains at zero throughout training means the constraint has no effect, whatever the loss curve shows.

6 Decision-time CVaR filter

Scaling the dual made the penalty bind in training but did not keep the portfolio within budget out of sample. A3 breached in 75.1% of test weeks, against 7.9% for A2. With the dual scaled, 27 of 30 checkpoints met the validation budget (8 of 30 for A2), so selection kept later and more return-seeking checkpoints (update 1010 on average, against 370). Four pre-specified rules for setting d from validation data all gave 71 to 78% breach. Classical rules scored against the same rolling budget reached 7.9% (minimum variance, inverse volatility and the min-CVaR LP), so the budget was feasible. Arms A5 to A8 repeat the setup and change only the named components (Table 1, Fig. 5).

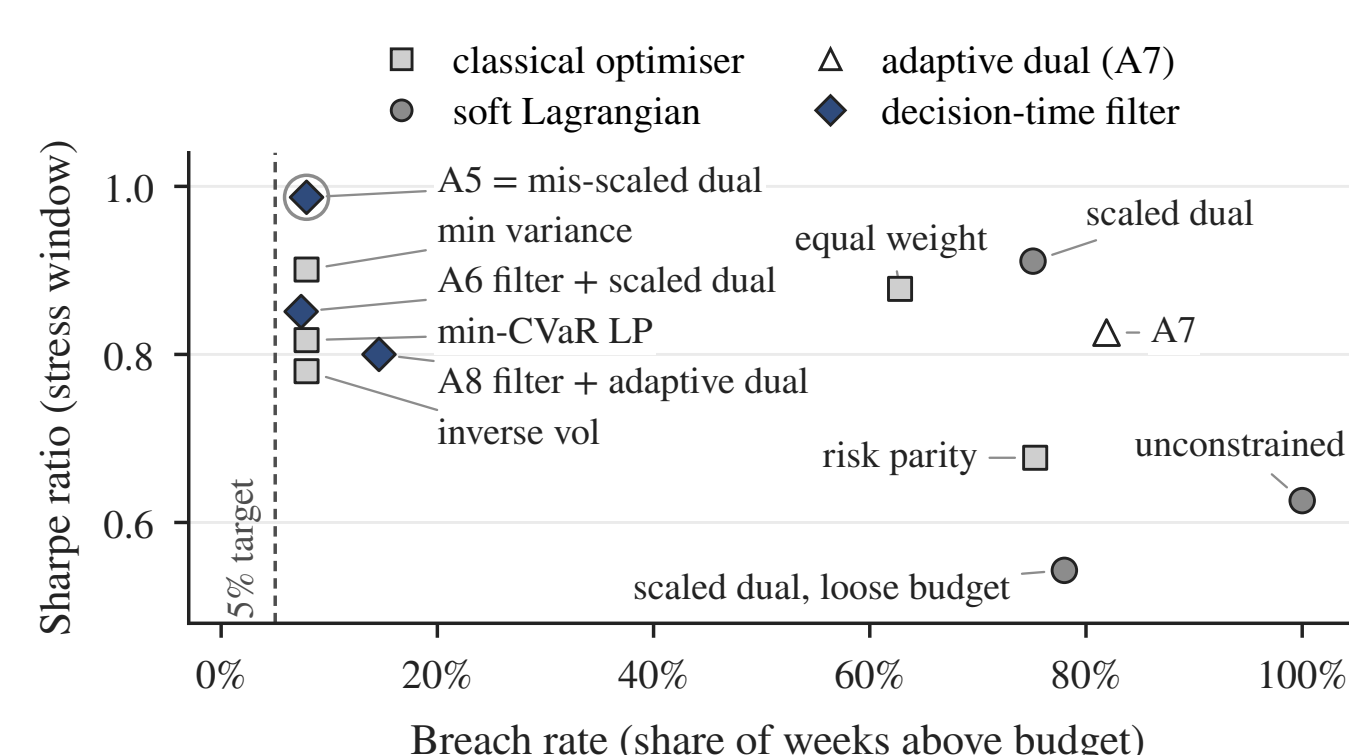


Figure 5. Breach rate against Sharpe on the stress window. The dashed line marks the 5% tolerance.

A6 lowered breach from 75.1% to 7.4% at $\eta_{\lambda} = 5.0$. In A5 the filter never triggered (0 of 1285 test weeks) because the mis-scaled policy was already within budget. A7 updated λ online from realised losses without the filter and still breached in 81.9% of weeks as λ rose to 7.69, which attributes the improvement in A6 to the filter. A8 reached 14.6%, with a range of 7 to 33% across seeds. The filtered weights stayed measurably different from minimum variance (A5 mean L_1 distance 0.476, correlation 0.786).

7 Stress-window and walk-forward results

Table 2. Stress window, mean over 5 seeds.

metric	unconstr.	constr.	change
Sharpe	0.63 ± 0.09	0.88 ± 0.04	+40%
Sortino	0.61	0.87	+44%
Calmar	0.32	0.48	+50%
max drawdown	0.206	0.118	-43%
CVaR ₉₅	0.0333	0.0189	-43%
CVaR ₉₉	0.0647	0.0327	-49%
violations per run	7.2	0.0	-100%

Table 3. CVaR₉₉ over 26-week walk-forward folds.

universe	folds	improved	Wilcoxon p
macro ETF (7)	20	9 (11 tied)	0.0077
sector ETF (10)	20	17	0.0003
pooled	40	26 of 27	8.8×10^{-6}
calendar clusters	20		3.3×10^{-4}

On the stress window, a paired block bootstrap of the CVaR₉₉ difference (2000 resamples, block length 4) gave -0.033 with 95% BCa interval $[-0.035, -0.022]$. On the sector universe the stress result replicated, with CVaR₉₉ 30% lower at unchanged Sharpe (0.752 against 0.748). In the walk-forward, a tie is a fold in which the budget never binds and the two learners coincide. Benjamini-Hochberg control [9] rejected the null on both universes, and the exact sign-flip test on calendar clusters gave $p = 3.8 \times 10^{-5}$. The Probabilistic Sharpe Ratio of the constrained allocator was at least 0.99 on both universes, but its Sharpe advantage over classical baselines did not survive the Deflated Sharpe Ratio [7].

8 Regimes and baselines

In the 10-fold macro walk-forward, the constrained policy did best relative to minimum variance in high-volatility weeks ($n = 71$, Sharpe 2.169 against 1.668) and worst in selloffs ($n = 61$, -0.809 against 0.035). Its CVaR₉₉ was below the unconstrained learner's in every regime (-29.1% calm, -20.7% high volatility, -27.4% selloff). A hybrid that applies constrained RL in high-volatility weeks and minimum variance otherwise reached Sharpe 1.316 (95% CI [0.66, 2.03]) against 1.209 ([0.55, 1.93]), and the intervals overlap. It beat minimum variance in 10 of 15 threshold settings. Tuned PPO (36 configurations) and SAC (8) did not generalise on 887 weekly observations. The selected PPO reached Sharpe 0.67 with CVaR₉₉ 0.070 on the stress window, and SAC 0.39.

9 Limitations and conclusion

The learned allocator did not beat rolling minimum variance on Sharpe out of sample, and the hybrid's gain came only from high-volatility weeks. The study uses weekly data with proportional costs and no market impact, the protocol reversal was shown on one 31-asset window, and the Gaussian forecast in the filter can understate fat-tailed losses. Within these limits, the constraint reduced tail risk in a way that survived multiple-testing correction, and the decision-time filter achieved the out-of-sample compliance that neither rescaling the dual nor recalibrating the budget achieved. Practitioners can detect the first failure by plotting λ during training and the second by evaluating on rolling folds.

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